

Phase

1

2

Task
1. Project plan
<i>Complete and submit a Project Plan</i>
2. Set up Raspberry Pi as Server and confirmed communication
<i>Set up Raspberry Pi as a server</i>
<i>Confirm communication between client computer and server (HTTP/SSH)</i>
3. STM32 Programming via HAL
<i>Explain common CAN messaging protocol (see document)</i>
<i>Implement the CAN messaging protocol by properly setting the filters (code demo) on STM32</i>
<i>Implement Wiring on STM32, push buttons/LED indicators (possible COVID restrictions)</i>
<i>Able to read CAN messages placed on the CAN bus (Raspberry Pi) and detect up/down request, floor request, door open/close, location of elevator car</i>
<i>Develop and implement a test plan to confirm functionality of software/hardware</i>
3. Controls / indicators (virtual and buttons)
<i>Controls (up/down) built in HTML (possibly add CSS if students work ahead)</i>
<i>Elevator Car Controls (up/down/floor number) built in HTML (possibly CSS if students work ahead)</i>
4. Finite State Machine (recommended)
<i>Develop Finite State Machine Logic for Elevator Operation</i>
1. User Interface / Client Program
<i>Fully working and completed front end Graphical User Interface /Webpage to control elevator</i>
<i>User action buttons/icons implemented</i>
<i>Indicators/lights etc. implemented</i>
<i>User action buttons developed to make changes to UI that are reflected in the database</i>
<i>User indicator buttons change - changes in database are reflected in UI</i>
2. Database Repository
<i>Database set up and accepting values/commands from website</i>
<i>Database has fields for logging information placed on the CAN bus</i>
3. Diagnostic Reports
<i>Produced based on content in database</i>
<i>Displayed on webpage</i>
4. Server Program
<i>Users are authorized/authenticated based on username/password</i>
<i>Server program provides logging data/accepts commands only from authorized users</i>
5. Announce floor numbers (Function of the Client or Server Program) for users with Visual Impairment

	<i>Implement a Server Program for announcing the floor numbers</i>
	<i>Speaker on elevator used to announce the current/next floor (speaker already connected to system)</i>
6. Sabbath mode (Function of the Client Program)	
	<i>Users may operate the elevator without having to push buttons (e.g. verbally, based on schedule/timer)</i>
1. CAN/LAN network communication	
	<i>Research and Implement a CAN/LAN communication bridge</i>
	<i>Ensure activity in CAN network changes values in database</i>
	<i>Changes in User Interface/Website changes value in database and generate CAN messages</i>
	<i>CAN Messages result in changes to database and GUI/Indicators</i>
	<i>Commands from CAN/LAN networks are sent to the controller and move the elevator / indicate floor</i>
2. Controller Software Written	
	<i>CAN Supervisor (Raspberry Pi) relays messages between CAN/LAN Network (see sample program on GitHub)</i>
3. Finite State Machine logic written	
	<i>Develop and Implement a finite state machine (using C/C++, PHP, Python to implement logic)</i>
4. Additional capability that shows ingenuity	
5. Ideas for Capstone (at least 3 per student)	
6. Final Presentation & Demonstration	

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Nigel/Ryan		
Nigel/Ryan		
Fergus/Ryan		
ALL		
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